

1. If the ratio of the fifth term from the beginning to the fifth term from the end in the expansion of $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}}\right)^n$ is $\sqrt{6} : 1$, then the third term from the beginning is:
[2023 (06 Apr Shift 1)]
(1) $30\sqrt{2}$
(2) $30\sqrt{3}$
(3) $60\sqrt{2}$
(4) $60\sqrt{3}$
2. If ${}^{2n}C_3 : {}^nC_3 = 10 : 1$, then the ratio $(n^2 + 3n) : (n^2 - 3n + 4)$ is
[2023 (06 Apr Shift 1)]
(1) 35 : 16
(2) 27 : 11
(3) 65 : 37
(4) 2 : 1
3. The coefficient of x^{18} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is _____
[2023 (06 Apr Shift 1)]
4. If the coefficients of x^7 in $\left(ax^2 + \frac{1}{2bx}\right)^{11}$ and x^{-7} in $\left(ax - \frac{1}{3bx^2}\right)^{11}$ are equal, then
[2023 (06 Apr Shift 2)]
(1) $729ab = 32$
(2) $32ab = 729$
(3) $64ab = 243$
(4) $243ab = 64$
5. Among the statements :
(S1) : $2023^{2022} - 1999^{2022}$ is divisible by 8.
(S2) : $13(13)^n - 11n - 13$ is divisible by 144 for infinitely many $n \in \mathbb{N}$
[2023 (06 Apr Shift 2)]
(1) Only (S2) is correct
(2) Only (S1) is correct
(3) Both (S1) and (S2) are correct
(4) Both (S1) and (S2) are incorrect
6. If the coefficients of three consecutive terms in the expansion of $(1+x)^n$ are the ratio 1 : 5 : 20 then the coefficient of the fourth term is
[2023 (08 Apr Shift 1)]
(1) 2436
(2) 5481
(3) 1827
(4) 3654
7. Let $[t]$ denote the greatest integer $\leq t$. If the constant term in the expansion of $\left(3x^2 - \frac{1}{2x^5}\right)^7$ is α then $[\alpha]$ is equal to _____
[2023 (08 Apr Shift 1)]
8. If a_α is the greatest term in the sequence $a_n = \frac{n^3}{n^4 + 147}$, $n = 1, 2, 3, \dots$, then α is equal to _____
[2023 (08 Apr Shift 1)]
9. The largest natural number n such that $3n$ divides $66!$ is _____
[2023 (08 Apr Shift 1)]
10. The absolute difference of the coefficients of x^{10} and x^7 in the expansion of $\left(2x^2 + \frac{1}{2x}\right)^{11}$ is equal to
[2023 (08 Apr Shift 2)]
(1) $13^3 - 13$
(2) $11^3 - 11$
(3) $10^3 - 10$
(4) $12^3 - 12$

11. $25^{190} - 19^{190} - 8^{190} + 2^{190}$ is divisible by
[2023 (08 Apr Shift 2)]
(1) neither 14 nor 34
(2) 14 but not by 34
(3) 34 but not by 14
(4) both 14 and 34
12. If the coefficient of x^7 in $\left(ax - \frac{1}{bx^2}\right)^{13}$ and the coefficient of x^{-5} in $\left(ax + \frac{1}{bx^2}\right)^{13}$ are equal, then a^4b^4 is equal to:
[2023 (10 Apr Shift 1)]
(1) 11
(2) 44
(3) 22
(4) 33.
13. The coefficient of x^7 in $(1 - x + 2x^3)^{10}$ is _____.
[2023 (10 Apr Shift 1)]
14. Let the number $(22)^{2022} + (2022)^{22}$ leave the remainder α when divided by 3 and β when divided by 7. Then $(\alpha^2 + \beta^2)$ is equal to
[2023 (10 Apr Shift 2)]
(1) 20
(2) 13
(3) 5
(4) 10
15. If the coefficients of x and x^2 in $(1+x)^p(1-x)^q$ are 4 and -5 respectively, then $2p+3q$ is equal to
[2023 (10 Apr Shift 2)]
(1) 60
(2) 69
(3) 66
(4) 63
16. The number of integral terms in the expansion of $\left(3^{\frac{1}{2}} + 5^{\frac{1}{4}}\right)^{680}$ is equal to
[2023 (11 Apr Shift 1)]
17. The mean of the coefficients of $x, x^2, \dots, x^7$ in the binomial expression of $(2+x)^9$ is _____.
[2023 (11 Apr Shift 1)]
18. If the 1011^{th} term from the end in the binomial expansion of $\left(\frac{4x}{5} - \frac{5}{2x}\right)^{2022}$ is 1024 times 1011^{th} term from the beginning, then $32|x|$ is equal to
[2023 (11 Apr Shift 2)]
(1) 15

(2) 10
(3) 12
(4) 8
19. The sum of the coefficients of three consecutive terms in the binomial expansion of $(1+x)^{n+2}$, which are in the ratio 1 : 3 : 5, is equal to
[2023 (11 Apr Shift 2)]
(1) 92
(2) 63
(3) 41
(4) 25
20. If $\frac{1}{n+1} {}^nC_n + \frac{1}{n} {}^nC_{n-1} + \dots + \frac{1}{2} {}^nC_1 + {}^nC_0 = \frac{1023}{10}$ then n is equal to
[2023 (12 Apr Shift 1)]
(1) 9
(2) 8
(3) 7
(4) 6

21. The sum, of the coefficients of the first 50 terms in the binomial expansion of $(1-x)^{100}$, is equal to
[2023 (12 Apr Shift 1)]
(1) $^{101}C_{50}$
(2) $^{99}C_{49}$
(3) $^{-101}C_{50}$
(4) $^{-99}C_{49}$
22. Fractional part of the number $\frac{4^{2022}}{15}$ is equal to
[2023 (13 Apr Shift 1)]
(1) $\frac{8}{15}$
(2) $\frac{4}{15}$
(3) $\frac{14}{15}$
(4) $\frac{1}{15}$
23. Let α be the constant term in the binomial expansion of $\left(\sqrt{x} - \frac{6}{x^2}\right)^n$, $n \leq 15$. If the sum of the coefficients of the remaining terms in the expansion is 649 and the coefficient of x^{-n} is $\lambda\alpha$, then λ is equal to -----
[2023 (13 Apr Shift 1)]
24. Let for $x \in \mathbb{R}$, $S_0(x) = x$, $S_k(x) = C_k x + k \int_0^x S_{k-1}(t) dt$ where $C_0 = 1$, $C_k = 1 - \int_0^1 S_{k-1}(x) dx$, $k = 1, 2, 3, \dots$. Then $S_2(3) + 6C_3$ is equal to -----
[2023 (13 Apr Shift 1)]
25. The coefficient of x^5 in the expansion of $\left(2x^3 - \frac{1}{3x^2}\right)^5$ is
[2023 (13 Apr Shift 2)]
(1) $\frac{80}{9}$
(2) 9
(3) 8
(4) $\frac{26}{3}$
26. The remainder, when 7^{103} is divided by 17, is
[2023 (13 Apr Shift 2)]
27. Let $(a + bx + cx^2)^{10} = \sum_{i=0}^{20} p_i x^i$, $a, b, c \in \mathbb{N}$. If $p_1 = 20$ and $p_2 = 210$, then $2(a + b + c)$ is equal to
[2023 (15 Apr Shift 1)]
(1) 6
(2) 15
(3) 12
(4) 8

ANSWER KEYS

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|------------|----------|-----------|---------|-----------|---------|-----------|-----------|
| 1. (4) | 2. (4) | 3. (5005) | 4. (1) | 5. (2) | 6. (4) | 7. (1275) | 8. (5) |
| 9. (31) | 10. (4) | 11. (3) | 12. (3) | 13. (960) | 14. (3) | 15. (4) | 16. (171) |
| 17. (2736) | 18. (2) | 19. (2) | 20. (1) | 21. (4) | 22. (4) | 23. (36) | 24. (18) |
| 25. (1) | 26. (12) | 27. (3) | | | | | |

1. (4)

Given expansion is $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}}\right)^n$

We know that general term of an expansion $(x + a)^n$ is $T_{r+1} = {}^nC_r x^{n-r} a^r$

(5th term from beginning)

$$T_{4+1} = {}^nC_4 \left(\sqrt[3]{2}\right)^{n-4} \cdot \left(\frac{1}{\sqrt[3]{3}}\right)^4$$

(5th term from end)

We know that r th term from the end in a binomial expansion is T_{n-r+2} from the beginning.

$$T'_5 = T_{n-4+1} = {}^nC_4 \left(\frac{1}{\sqrt[3]{3}}\right)^{n-4} \left(\sqrt[3]{2}\right)^4$$

It is also given that the ratio of the fifth term from the beginning to the fifth term from the end is $\sqrt{6} : 1$.

$$\text{Now, } \frac{T_5}{T'_5} = \frac{\sqrt{6}}{1}$$

$$\Rightarrow 2^{\frac{n-8}{4}} \cdot 3^{\frac{n-8}{4}} = \sqrt{6}$$

$$\therefore \left(\sqrt{6}\right)^{\frac{n-8}{2}} = \left(\sqrt{6}\right)$$

$$\therefore n = 10$$

$$\therefore T_3 = {}^{10}C_2 \left(\sqrt[3]{2}\right)^8 \left(\frac{1}{\sqrt[3]{3}}\right)^2 = \frac{45 \times 4}{\sqrt{3}}$$

$$= 60\sqrt{3}$$

Hence, this is the correct option.

2. (4)

We have been given that ${}^{2n}C_3 : {}^nC_3 = 10$

We know that ${}^nC_r = \frac{n!}{r!(n-r)!}$

$$\Rightarrow \frac{{}^{2n}C_3}{{}^nC_3} = 10$$

$$\Rightarrow \frac{\frac{(2n)!}{3!(2n-3)!}}{\frac{n!}{3!(n-3)!}} \times \frac{3!(n-3)!}{n!} = 10$$

$$\Rightarrow \frac{(2n)(2n-1)(2n-2)(2n-3)!}{(2n-3)!} \times \frac{(n-3)!}{n(n-1)(n-2)(n-3)!} = 10$$

$$\Rightarrow \frac{(2n)(2n-1)(2n-2)}{n(n-1)(n-2)} = 10$$

$$\Rightarrow \frac{(4)(2n-1)}{(n-2)} = 10$$

$$\Rightarrow 8n - 4 = 10n - 20$$

$$\Rightarrow n = 8$$

$$\text{Hence, the value of } \frac{n^2+3n}{n^2-3n+4} \text{ is } = \frac{8^2+3(8)}{8^2-3(8)+4} = \frac{88}{44} = 2$$

Therefore, the value of $\frac{n^2+3n}{n^2-3n+4}$ is 2 : 1.

3. (5005)

Given,

Binomial expression $\left(x^4 - \frac{1}{x^3}\right)^{15}$

Now general term is given by,

$$T_{r+1} = {}^{15}C_r (x^4)^{15-r} \left(-\frac{1}{x^3}\right)^r$$

$$\Rightarrow T_{r+1} = {}^{15}C_r x^{60-7r} (-1)^r$$

For coefficient of x^{18} , equate $60 - 7r = 18$

$$\Rightarrow r = 6$$

$$\text{Coefficient of } x^{18} = (-1)^6 \cdot {}^{15}C_6 = 5005$$

4. (1)

The coefficient of x^7 in $\left(ax^2 + \frac{1}{2bx}\right)^{11}$ and $\left(ax - \frac{1}{3bx^2}\right)^{11}$ are equal, then

$$T_{r+1} = {}^{11}C_r (ax^2)^{11-r} \left(\frac{1}{2bx}\right)^r$$

$$= {}^{11}C_r (a)^{11-r} \left(\frac{1}{2b}\right)^r x^{22-3r}$$

Now solving $22 - 3r = 7 \Rightarrow r = 5$

Coefficient of x^{-7} in $\left(ax - \frac{1}{3bx^2}\right)^{11}$

$$T_{r+1} = {}^{11}C_r (ax)^{11-r} \left(-\frac{1}{3bx^2}\right)^r$$

$$= {}^{11}C_r a^{11-r} \left(-\frac{1}{3b}\right)^r x^{11-3r}$$

Now solving $11 - 3r = -7 \Rightarrow r = 6$

Now equating coefficient of x^7 and x^{-7} we get,

$$\therefore {}^{11}C_5 (a)^6 \left(\frac{1}{2b}\right)^5 = {}^{11}C_6 a^5 \left(-\frac{1}{3b}\right)^6$$

$$\Rightarrow 3^6 ab = 32$$

$$\Rightarrow 729ab = 32$$

Hence this is the correct option.

5. (2)

Given,

(S1) : $(2023)^{2022} - (1999)^{2022}$ is divisible by 8

Now we know that $(x - y)$ divides $(x^n - y^n) \forall n \in \mathbb{N}$

So, $(2023 - 1999)$ divides $(2023)^{2022} - (1999)^{2022}$

$$\Rightarrow 24 \text{ divides } (2023)^{2022} - (1999)^{2022}$$

$$\Rightarrow 8 \text{ will divide } (2023)^{2022} - (1999)^{2022}$$

As 8 divides 24

Hence, (S1) is correct

Now solving,

(S2) : $13(13)^n - 11n - 13$ is divisible by 144 for $n \in \mathbb{N}$

So using binomial theorem in $(1 + 12)^n$ we get,

$$13(1 + 12)^n - 11n - 13$$

$$= 13({}^nC_0 + {}^nC_1 12 + {}^nC_2 12^2 + \dots + {}^nC_n 12^n) - 11n - 13$$

$$= 12 \times 13n - 11n + 12^2 \lambda$$

$$= 145n + 144\lambda \text{ which is not divisible by 144}$$

Hence, (S2) is incorrect

6. (4)

Given,

$${}^nC_{r-1} : {}^nC_r : {}^nC_{r+1} = 1 : 5 : 20$$

Now using the formula $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$ we get,

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} = \frac{5}{1}$$

$$\Rightarrow n - 6r + 1 = 0 \dots\dots\dots(1)$$

$$\text{And } \frac{{}^nC_{r+1}}{{}^nC_r} = \frac{n-r}{r+1} = \frac{20}{5}$$

$$\Rightarrow n - 5r - 4 = 0 \dots\dots\dots(2)$$

Solving above equations we get,

$$n = 29 \text{ \& } r = 5$$

$$\text{So, the coefficient of forth term will be } {}^nC_3 = {}^{29}C_3 = \frac{29 \times 28 \times 27}{3 \times 2 \times 1} = 3654$$

7. (1275)

The given expansion is $\left(3x^2 - \frac{1}{2x^5}\right)^7$.

The general term in the binomial expansion of $(x + a)^n$ is given by $T_{r+1} = {}^nC_r x^{n-r} a^r$.

$$\Rightarrow T_{r+1} = {}^7C_r (3x^2)^{7-r} \left(\frac{-1}{2x^5}\right)^r$$

$$= {}^7C_r (-1)^r \left(\frac{3^{7-r}}{2^r}\right) x^{14-2r-5r}$$

Now for term independent of $x \Rightarrow 14 - 2r - 5r = 0$

$$\Rightarrow r = 2$$

Coefficient of x^0 is ${}^7C_2 (-1)^2 \times \left(\frac{3^{7-2}}{2^2}\right)$

$$= \frac{7 \times 6}{2} \times \frac{3^5}{2^2}$$

$$\Rightarrow \alpha = \frac{5103}{4}$$

$$\Rightarrow [\alpha] = 1275$$

Hence this is the required answer.

8. (5)

Let,

$$y = \frac{x^3}{x^4+147} = f(x)$$

We know that, for increasing function $\frac{dy}{dx} > 0$

So, differentiating the given function $f(x) = \frac{x^3}{x^4+147}$ we get,

$$\frac{dy}{dx} = \frac{-x^2(x^4-441)}{(x^4+147)^2}$$

$$\Rightarrow \frac{-x^2(x^4-441)}{(x^4+147)^2} > 0$$

$$\Rightarrow \frac{x^2(x^4-441)}{(x^4+147)^2} < 0$$

$$\Rightarrow x^4 < 441$$

Now for maxima/minima $\frac{dy}{dx} = 0$

$$\Rightarrow x^4 = 441$$

$$\Rightarrow x = \alpha, 4 < \alpha < 5$$

$\Rightarrow$ Maximum value of $f(x)$ is at $x = 4$ or $x = 5$

$$f(4) = \frac{64}{403}, f(5) = \frac{125}{772}$$

$$\therefore f(5) > f(4)$$

$$\Rightarrow \alpha = 5$$

9. (31)

We know that,

Maximum value of n for p^n where p is prime divides the number $a!$ is given by,

$$\left[\frac{a}{p}\right] + \left[\frac{a}{p^2}\right] + \left[\frac{a}{p^3}\right] + \left[\frac{a}{p^4}\right] + \dots$$

Since, 3 is a prime number, so we have

$$= \left[\frac{66}{3}\right] + \left[\frac{66}{3^2}\right] + \left[\frac{66}{3^3}\right] + \left[\frac{66}{3^4}\right] + \dots$$

$$= 22 + 7 + 2 + 0$$

$$= 31$$

So, maximum value of n is 31.

10. (4)

The given expansion is $\left(2x^2 + \frac{1}{2x}\right)^{11}$.

The general term in the binomial expansion of $(x + a)^n$ is given by $T_{r+1} = {}^nC_r x^{n-r} a^r$.

$$\Rightarrow T_{r+1} = {}^{11}C_r (2x^2)^{11-r} \left(\frac{1}{2x}\right)^r$$

$$= {}^{11}C_r (2^{11-r-r}) x^{22-2r-r}$$

Now for coefficient of $x^7 \Rightarrow 22 - 3r = 7$

$$\Rightarrow r = 5$$

Coefficient of x^7 is ${}^{11}C_5 (2^{11-5 \times 2})$

Now for coefficient of $x^{10} \Rightarrow 22 - 3r = 10$

$$\Rightarrow r = 4$$

Coefficient of x^{10} is ${}^{11}C_4 (2^{11-2 \times 4})$

We need the absolute difference, we get it by $|{}^{11}C_5 (2^{11-5 \times 2}) - {}^{11}C_4 (2^{11-4 \times 2})|$.

$$= |924 - 2640| = 1716$$

$$= 12^3 - 12$$

Hence the absolute difference is $= 12^3 - 12$.

11. (3)

Given,

$$25^{190} - 19^{190} = 8^{190} + 2^{190}$$

Now $(25^{190} - 19^{190})$, 8^{190} , 2^{190} are divisible by 2, as 25^{190} will give remainder as 1 and 19^{190} will give remainder as 1 when divided by 2, so combined $25^{190} - 19^{190}$ will give remainder 0

Now, $(25^{190} - 8^{190})$ & $(-19^{190} + 2^{190})$ are divisible by 17, $a^n - b^n$ is divisible by $a - b$ if n is even,

So given number is divisible by 34 as it is divisible by 2 & 17 both,

Now, $25^{190} = (7 \times 3 + 4)^{190}$, dividing by 7 we get,

$$(7 \times 3 + 4)^{190} = 4^{190} = 16^{95} = 2^{95} = 4 \cdot 8^{31}$$

Now when $4 \cdot (7 + 1)^{31}$ divided by 7 will give remainder as $4 \cdot 1 = 4$

And $19^{190} = (7 \times 3 - 2)^{190}$, when divided by 7 we get,

$$(7 \times 3 - 2)^{190} = (-2)^{190} = 2^{190} = 2 \times 8^{63} = 2 \times (7 + 1)^{63}$$

So, when $2 \times (7 + 1)^{63}$ divided by 7 will give remainder as 2

Similarly $8^{190} = (7 + 1)^{190}$ will give 1 remainder

And $2^{190} = 2(7 + 1)^{63}$ will give 1 remainder,

So combining all we get, $25^{190} - 19^{190} - 8^{190} + 2^{190} = 4 - 2 - 1 + 1 = 5$

$\Rightarrow$ Not divisible by 7, so it will not be divisible by 14

12. (3)

Given,

The coefficient of x^7 in the expansion of $\left(ax^2 - \frac{1}{bx}\right)^{13}$ is equal to the coefficient of x^{-5} in $\left(ax + \frac{1}{bx^2}\right)^{13}$.

We know that, the general term T_{r+1} in the expansion $(a+b)^n$ is

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Applying to $\left(ax - \frac{1}{bx^2}\right)^{13}$, we get

$$T_{r+1} = {}^{13}C_r (ax)^{13-r} \left(-\frac{1}{bx^2}\right)^r$$

$$\Rightarrow T_{r+1} = (-1)^r \times {}^{13}C_r (a)^{13-r} (x)^{13-3r} (b)^{-r}$$

Therefore, $13 - 3r = 7 \Rightarrow r = 2$ for coefficient of x^7 .

Thus,

$$T_3 = {}^{13}C_2 \left(\frac{a^{11}}{b^2}\right)$$

Similarly, applying to $\left(ax + \frac{1}{bx^2}\right)^{13}$, we get

$$T_{r+1} = {}^{13}C_r (ax)^{13-r} \left(\frac{1}{bx^2}\right)^r$$

$$\Rightarrow T_{r+1} = {}^{13}C_r (a)^{13-r} (x)^{13-3r} (b)^{-r}$$

Therefore, $13 - 3r = -5$ for coefficient of x^{-5}

$$\Rightarrow r = 6$$

So,

$$T_7 = {}^{13}C_6 (a)^7 (b)^{-6}$$

Hence, applying the given condition we get

$${}^{13}C_2 \left(\frac{a^{11}}{b^2}\right) = {}^{13}C_6 (a)^7 (b)^{-6}$$

$$\Rightarrow a^4 b^4 = \frac{{}^{13}C_6}{{}^{13}C_2}$$

$$\Rightarrow a^4 b^4 = \frac{13!}{7!6!} \times \frac{2!11!}{13!}$$

$$\Rightarrow a^4 b^4 = \frac{11 \times 10 \times 9 \times 8}{6 \times 5 \times 4 \times 3}$$

$$\Rightarrow a^4 b^4 = 22$$

13. (960)

We need to find the coefficient of x^7 in $(1 - x + 2x^3)^{10}$.

We know that for $(x + y + z)^n$,

$$T_n = \frac{n!}{a!b!c!} (x)^a (y)^b (z)^c \text{ such that } a + b + c = n$$

Now for $(1 - x + 2x^3)^{10}$

$$T_n = \frac{10!}{a!b!c!} (1)^a (-x)^b (2x^3)^c$$

$$= \frac{10!}{a!b!c!} (-1)^b 2^c x^{b+3c} \text{ with } a + b + c = 10$$

Here we need coefficient of x^7 .

Hence the combinations would be

a	b	c
3	7	0
5	4	1
7	1	2

which satisfies both $a + b + c = 10$ and $b + 3c = 7$.

Hence, coefficient of x^7 will be,

$$= \frac{10!}{2!1!7!} (2)^2 (-1)^1$$

$$+ \frac{10!}{1!4!5!} (2)^1 (-1)^4 + \frac{10!}{0!7!3!} (2)^0 (-1)^7$$

$$= 960$$

14. (3)

Given that α be the remainder when $(22)^{2022} + (2022)^{22}$ is divided by 3 and β be the remainder when the same is divided by 7.

$$\Rightarrow (22)^{2022} + (2022)^{22} = (21+1)^{2022} + (2022)^{22}$$

Here $(2022)^{22}$ is divisible by 3 as 2022 is divisible by 3.

So on expanding $(21+1)^{2022}$, we get

$$\Rightarrow (21+1)^{2022} = {}^{2022}C_0(21)^{2022} + {}^{2022}C_1(21)^{2021} + \dots + {}^{2022}C_{2022}(1)^{2022}$$

$$= 3(3^{2021} \times 7^{2022} + {}^{2022}C_1 \times 3^{2020} \times 7^{2021} + \dots) + 1$$

$$= 3k_1 + 1$$

In this case the remainder is 1

$$\text{Hence, } \alpha = 1$$

$$\text{Now, } \Rightarrow (22)^{2022} + (2022)^{22} = (21+1)^{2022} + (2023-1)^{22}$$

$$\text{Take } (2023-1)^{22}$$

$$\Rightarrow (2023-1)^{22} = {}^{22}C_0(2023)^{22} - {}^{22}C_1(2023)^{21} + \dots + {}^{22}C_{22}(-1)^{22}$$

$$= 7({}^{22}C_0(7)^{21}(289)^{22} - {}^{22}C_1(7)^{20}(289)^{21} + \dots) + 1$$

$$= 7k_2 + 1$$

$$\Rightarrow (21+1)^{2022} + (2023-1)^{22} = 7k_1 + 1 + 7k_2 + 1$$

$$= 7\mu + 2$$

$$\Rightarrow \beta = 2.$$

$$\text{Hence, } \alpha^2 + \beta^2 = 1^2 + 2^2 = 5.$$

Therefore, the required answer is 5.

15. (4)

Given that The coefficient of x and x^2 in $(1+x)^p(1-x)^q$ are 4 and -5 .

$$\Rightarrow (1+x)^p(1-x)^q = \left(1+px+\frac{p(p-1)}{2!}x^2+\dots\right)\left(1-qx+\frac{q(q-1)}{2!}x^2-\dots\right)$$

Now coefficient of x from the above expansion will be

$p-q$ which is equal to 4

$$\Rightarrow p-q=4.$$

Similarly coefficient of x^2 is -5 .

$$\Rightarrow \frac{p(p-1)}{2} + \frac{q(q-1)}{2} - pq = -5$$

$$\Rightarrow \frac{p^2-2pq+q^2}{2} - \frac{(p+q)}{2} = -5$$

$$\Rightarrow \frac{(p-q)^2}{2} - \frac{(p+q)}{2} = -5$$

$$\Rightarrow \frac{16}{2} + 5 = \frac{(p+q)}{2}$$

$$\Rightarrow p+q=26 \text{ and } p-q=4$$

On solving the above equations we get,

$$p=15 \text{ and } q=11.$$

$$\Rightarrow 2p+3q=2(15)+3(11)=63$$

Therefore, the required answer is 63.

16. (171)

Given expansion is $\left(3^{\frac{1}{2}} + 5^{\frac{1}{4}}\right)^{680}$.

The general term in the expansion $(x+a)^n$ is $T_{r+1} = {}^nC_r \cdot x^{n-r} \cdot a^r$

$$\Rightarrow T_{r+1} = {}^{680}C_r \left(3^{\frac{1}{2}}\right)^{680-r} \left(5^{\frac{1}{4}}\right)^r, 0 \leq r \leq 680$$

$$= {}^{680}C_r (3)^{\frac{680-r}{2}} (5)^{\frac{r}{4}}$$

For $(3)^{\frac{680-r}{2}} (5)^{\frac{r}{4}}$ to be rational r should be a multiple of 4.

$$\Rightarrow r = 0, 4, 8, 12, \dots, 680$$

$$\Rightarrow a_n = a + (n-1)d$$

$$\Rightarrow 680 = 0 + (n-1)4$$

$$\Rightarrow n = 171$$

That means r can take 171 values.

Hence, the required answer is 171.

17. (2736)

We know that.

Binomial expansion of $(2+x)^9$ is given by,

$$(2+x)^9 = {}^9C_0 2^9 + {}^9C_1 x 2^8 + \dots + {}^9C_9 x^9 2^0$$

Now put $x = 1$ we get,

$$(2+1)^9 = {}^9C_0 2^9 + {}^9C_1 2^8 + {}^9C_2 2^7 + \dots + {}^9C_9 2^0$$

$$\Rightarrow 3^9 = {}^9C_0 2^9 + {}^9C_1 2^8 + {}^9C_2 2^7 + \dots + {}^9C_9 2^0$$

So, sum of coefficient of $x, x^2, x^3, \dots, x^7$ will be,

$$\Rightarrow {}^9C_1 2^8 + {}^9C_2 2^7 + \dots + {}^9C_7 2^2 = 3^9 - {}^9C_0 2^9 - {}^9C_8 2^1 - {}^9C_9 2^0$$

$$\Rightarrow {}^9C_1 2^8 + {}^9C_2 2^7 + \dots + {}^9C_7 2^2 = 3^9 - 2^9 - 18 - 1$$

$$\Rightarrow {}^9C_1 2^8 + {}^9C_2 2^7 + \dots + {}^9C_7 2^2 = 19683 - 512 - 19$$

$$\Rightarrow {}^9C_1 2^8 + {}^9C_2 2^7 + \dots + {}^9C_7 2^2 = 19152$$

$$\text{Now mean is given by } \frac{19152}{7} = 2736$$

18. (2)

Given,

The 1011^{th} term from the end in the binomial expansion of $\left(\frac{4x}{5} - \frac{5}{2x}\right)^{2022}$ is 1024 times 1011^{th} term from the beginning,

So, according to question we have, $\frac{T_{1013}}{T_{1011}} = 1024$

$$\Rightarrow \frac{{}^{2022}C_{1012} \left(\frac{4x}{5}\right)^{1010} \cdot \left(-\frac{5}{2x}\right)^{1012}}{{}^{2022}C_{1010} \left(\frac{4x}{5}\right)^{1012} \left(-\frac{5}{2x}\right)^{1010}} = 1024$$

$$\Rightarrow \frac{\left(\frac{-5}{2x}\right)^2}{\left(\frac{4x}{5}\right)^2} = 1024$$

$$\Rightarrow \frac{5^4}{2^6 x^4} = 2^{10}$$

$$\Rightarrow \left(\frac{5}{x}\right)^4 = 2^{16}$$

$$\Rightarrow |x| = \frac{5}{16}$$

$$\Rightarrow 32|x| = 10$$

Note this question was bonus in Jee Main 2023 April session as answer key was not matching, so we have modified the question.

19. (2)

Ratio of three consecutive terms in the binomial expansion of $(1+x)^{n+2}$ is

$${}^{n+2}C_{r-1} : {}^{n+2}C_r : {}^{n+2}C_{r+1} :: 1 : 3 : 5$$

So,

$$\Rightarrow \frac{{}^{n+2}C_r}{{}^{n+2}C_{r-1}} = \frac{3}{1}$$

$$\Rightarrow \frac{n-r+3}{r} = 3$$

$$\Rightarrow n+3 = 4r \dots (1)$$

And,

$$\frac{{}^{n+2}C_{r+1}}{{}^{n+2}C_r} = \frac{5}{3}$$

$$\Rightarrow \frac{n+2-r}{r+1} = \frac{5}{3}$$

$$\Rightarrow 3n+6-3r = 5r+5$$

$$\Rightarrow 3n+1 = 8r \dots (2)$$

$$\text{So, } 3(4r-3)+1 = 8r$$

$$\Rightarrow 12r-9+1 = 8r$$

$$\Rightarrow 4r = 8$$

$$\Rightarrow r = 2$$

$$\text{So, } n = 5$$

Hence, we have

$$(1+x)^{n+2} = (1+x)^7$$

Sum of consecutive terms is

$${}^7C_1 + {}^7C_2 + {}^7C_3$$

$$= 7 + 21 + 35 = 63$$

20. (1)

$$\text{Given that } \frac{1}{n+1} {}^nC_n + \frac{1}{n} {}^nC_{n-1} + \dots + \frac{1}{2} {}^nC_1 + {}^nC_0 = \frac{1023}{10}$$

$$\Rightarrow \sum_{r=0}^n \frac{1}{r+1} {}^nC_r = \frac{1023}{10} \left(\therefore {}^{n+1}C_{r+1} = \frac{n+1}{r+1} {}^nC_r \right)$$

$$\Rightarrow \sum_{r=0}^n \frac{1}{n+1} {}^{n+1}C_{r+1} = \frac{1023}{10}$$

$$\Rightarrow \frac{1}{n+1} [{}^{n+1}C_1 + {}^{n+1}C_2 + \dots + {}^{n+1}C_{n+1}] = \frac{1023}{10}$$

$$\text{We know that } {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$$

$$\Rightarrow \frac{2^{n+1}-1}{n+1} = \frac{1023}{10} = \frac{2^{10}-1}{10}$$

$$\Rightarrow n+1 = 10$$

$$\Rightarrow n = 9$$

Hence this is the correct option.

21. (4)

$$\text{We know that } (1-x)^{100} = {}^{100}C_0 - {}^{100}C_1 \cdot x + {}^{100}C_2 \cdot x^2 - \dots - {}^{100}C_{49} \cdot x^{49} + \dots + {}^{100}C_{100} \cdot x^{100}$$

Now let, the sum of first fifty coefficients be S .

$$\Rightarrow S = {}^{100}C_0 - {}^{100}C_1 + {}^{100}C_2 - \dots - {}^{100}C_{49}$$

Let us substitute $x = 1$ in the expansion.

$$\Rightarrow (1-1)^{100} = {}^{100}C_0 - {}^{100}C_1 \cdot 1 + {}^{100}C_2 \cdot 1^2 - \dots - {}^{100}C_{49} \cdot 1^{49} + \dots + {}^{100}C_{100} \cdot 1^{100}$$

$$\Rightarrow 0 = {}^{100}C_0 - {}^{100}C_1 + {}^{100}C_2 - \dots - {}^{100}C_{49} + {}^{100}C_{50} - {}^{100}C_{51} + {}^{100}C_{52} - \dots + {}^{100}C_{100}$$

There are total of 101 terms and the middle term is $T_{51} = {}^{100}C_{50}$.

$$\text{As } {}^nC_r = {}^nC_{n-r}$$

$$\Rightarrow 0 = {}^{100}C_0 - {}^{100}C_1 + {}^{100}C_2 - \dots - {}^{100}C_{49} + {}^{100}C_{50} - {}^{100}C_{49} + {}^{100}C_{48} - \dots + {}^{100}C_0$$

$$\Rightarrow 0 = {}^{100}C_{50} + {}^{100}C_0 - {}^{100}C_1 + {}^{100}C_2 - \dots - {}^{100}C_{49} - {}^{100}C_{49} + {}^{100}C_{48} - \dots + {}^{100}C_0$$

$$\Rightarrow 0 = {}^{100}C_{50} + 2({}^{100}C_0 - {}^{100}C_1 + {}^{100}C_2 - \dots - {}^{100}C_{49})$$

$$\Rightarrow -\frac{{}^{100}C_{50}}{2} = ({}^{100}C_0 - {}^{100}C_1 + {}^{100}C_2 - \dots - {}^{100}C_{49})$$

$$\Rightarrow {}^{100}C_0 - {}^{100}C_1 + {}^{100}C_2 - \dots - {}^{100}C_{49} = -\frac{{}^{100}C_{50}}{2}$$

$$\text{Therefore, the required answer is } -\frac{{}^{100}C_{50}}{2} = -\frac{1}{2} \times \frac{100}{50} {}^{99}C_{49} = -{}^{99}C_{49}.$$

22. (4)

$$\text{Given that } \frac{4^{2022}}{15}$$

$$\text{We know that } (a+x)^n = {}^nC_0 a^n x^0 + {}^nC_1 a^{n-1} x^1 + \dots + {}^nC_n a^0 x^n$$

$$\Rightarrow \frac{4^{2022}}{15} = \frac{16^{1011}}{15} = \frac{(15+1)^{1011}}{15}$$

$$= \frac{{}^{1011}C_0 (15)^{1011} + {}^{1011}C_1 (15)^{1010} + \dots + {}^{1011}C_{1011} (15)^0}{15}$$

$$= \frac{{}^{1011}C_0 (15)^{1011} + {}^{1011}C_1 (15)^{1010} + \dots + {}^{1011}C_{1010} (15)^1}{15} + \frac{1}{15}$$

$$\therefore \text{Fractional part of } \frac{4^{2022}}{15} = \frac{1}{15}$$

Hence this is the required option.

23. (36)

Given,

$$\left(\sqrt{x} - \frac{6}{x^{\frac{3}{2}}}\right)^n, n \leq 15$$

Now r^{th} term is given by

$$T_{r+1} = {}^nC_r \left(x^{\frac{1}{2}}\right)^{n-r} \left(-6x^{-\frac{3}{2}}\right)^r$$

$$\Rightarrow T_{r+1} = (-6)^r \cdot {}^nC_r \cdot x^{\frac{n-r}{2}} \cdot x^{-\frac{3r}{2}}$$

$$\Rightarrow T_{r+1} = (-6)^r \cdot {}^nC_r \cdot x^{\frac{n-4r}{2}}$$

Now we have to take, $n = 4r$ for constant term so n is divisible by 4.

$$\text{So, } \alpha = {}^nC_{\frac{n}{4}} (-6)^{\frac{n}{4}}$$

$$\text{And sum of all coefficients will be } \left(\sqrt{1} - \frac{6}{1^{\frac{3}{2}}}\right)^n = (-5)^n$$

$$\text{Also given, } (-5)^n - {}^nC_{\frac{n}{4}} (-6)^{\frac{n}{4}} = 649$$

Now by hit and trial we get, $n = 4$

$$\text{Hence, } \alpha = {}^4C_1 (-6)^{\frac{4}{4}} = -24$$

Now for coefficient of x^{-4}

$$\frac{n-4r}{2} = -4$$

$$\Rightarrow n = 4r - 8 \Rightarrow \boxed{r=3}$$

$$\text{So, } \lambda(\alpha) = (-6)^3 \cdot {}^4C_3$$

$$\Rightarrow \lambda(-24) = (-6)^3 \cdot {}^4C_3$$

$$\Rightarrow \lambda = 36$$

24. (18)

Given,

$$S_0(x) = x, S_k(x) = C_k x + k \int_0^x S_{k-1}(t) dt$$

Now for $S_0(x) = x, C_0 = 1$

Now solving, $C_k = 1 - \int_0^1 S_{k-1}(x) dx$ for $k = 1$ we get,

$$C_1 = 1 - \int_0^1 x dx = \frac{1}{2}$$

Now putting $k = 1$ in $S_k(x) = C_k x + k \int_0^x S_{k-1}(t) dt$ we get,

$$\Rightarrow S_1(x) = \frac{x}{2} + 1 \cdot \int_0^x t dt = \frac{x}{2} + \frac{x^2}{2}$$

Now for $k = 2$ we get,

$$C_2 = 1 - \int_0^1 \left(\frac{x}{2} + \frac{x^2}{2}\right) dx = \frac{7}{12}$$

$$\text{And } S_2(x) = \frac{7}{12}x + 2 \int_0^x \left(\frac{t}{2} + \frac{t^2}{2}\right) dt$$

$$\Rightarrow S_2(x) = \frac{7x}{12} + \frac{x^2}{2} + \frac{x^3}{3}$$

Now taking $k = 3$ we get,

$$C_3 = 1 - \int_0^1 \left(\frac{7x}{12} + \frac{x^2}{2} + \frac{x^3}{3}\right) dx = \frac{11}{24}$$

Now finding the value of $S_2(3) + 6 \cdot C_3$ we get,

$$S_2(3) + 6 \cdot C_3 = \frac{7 \times 3}{12} + \frac{3^2}{2} + \frac{3^3}{3} + 6 \times \frac{11}{24}$$

$$\Rightarrow S_2(3) + 6 \cdot C_3 = \frac{61}{4} + \frac{11}{4} = 18$$

25. (1)

General term of the binomial expansion of $\left(2x^3 - \frac{1}{3x^2}\right)^5$ is

$$T_{r+1} = {}^5C_r \left(2x^3\right)^{5-r} \left(\frac{-1}{3x^2}\right)^r$$

$$\Rightarrow T_{r+1} = {}^5C_r (2)^{5-r} \left(-\frac{1}{3}\right)^r x^{15-5r}$$

For coefficient of x^5 , we must have

$$15 - 5r = 5 \Rightarrow r = 2$$

So, required coefficient is

$$= {}^5C_2 (2)^3 \left(\frac{1}{9}\right)$$

$$= 10 \times 8 \times \frac{1}{9}$$

$$= \frac{80}{9}$$

26. (12)

Using the binomial expansion in $7^{103} = (7^2)^{51} \cdot 7 = (51 - 2)^{51} \cdot 7$ we get,

$$(51 - 2)^{51} \cdot 7 = 7 \cdot ({}^{51}C_0 51^{51} \cdot 2^0 - {}^{51}C_1 51^{50} \cdot 2^1 + \dots - {}^{51}C_{51} 2^{51})$$

Now 51 is divisible by 17, so when above equation is divided by 17 we get, $7 \cdot (-2^{51})$ as remainder,

Now again using binomial in $7 \cdot (-2^{51}) = -56(2^4)^{12} = -56(17 - 1)^{12}$ we get,

$$-56(17 - 1)^{12} = -56({}^{12}C_0 17^{12} \cdot 1^0 - {}^{12}C_1 17^{11} \cdot 1^1 + \dots + {}^{12}C_{12} 1^{12})$$

Now dividing above equation by 17 we get,

$$(-56 \times 1) \div 17 = -5,$$

Now changing negative remainder to positive we get, $-5 + 17 = 12$,

Hence, when 7^{103} divided by 17 gives 12 as remainder.

27. (3)

Given that $(a + bx + cx^2)^{10} = \sum_{i=0}^{20} p_i x^i$, $a, b, c \in \mathbb{N}$

General term : $\frac{10!}{r_1! r_2! r_3!} (a)^{r_1} (bx)^{r_2} (cx^2)^{r_3}$

For coefficient of x : $r_2 + 2r_3 = 1$

$$\text{Let } \begin{matrix} r_1 & r_2 & r_3 \\ 9 & 1 & 0 \end{matrix}$$

$$\therefore \text{coefficient of } x = \frac{10!}{9!} a^9 b^1 = 20$$

$$\Rightarrow a^9 \cdot b = 2 \dots \dots (i)$$

$$\text{Coefficient of } x^2 : \frac{10!}{8! 2! 0!} a^8 \cdot b^2 + \frac{10!}{9! 0! 1!} a^9 \cdot c = 210$$

$$\Rightarrow 45a^8 \cdot b^2 + 10 \cdot a^9 \cdot c = 210$$

$$\Rightarrow 9a^8 b^2 + 2a^9 \cdot c = 42 \dots \dots (ii)$$

As $a, b, c \in \mathbb{N}$, so by hit and trial in equation (i) & (ii) we get, $a = 1, b = 2, c = 3$

$$\text{Hence, } 2(a + b + c) = 2(3 + 2 + 1) = 12$$